

Automating cable installation

Robospool

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Abstract—

In this work we present our approach to solving an industrial challenge that involves the automated crimping of cables using a robot manipulator. The scenario we are trying to solve is the automated detection of the position and orientation of a cable, randomly placed on top of a table, and then its handling using a UR5e manipulator robot in order to correctly pick it and transport it with precision in a crimping machine hole. To do that, two different approaches in the detection of the cable has been used and compared, and a simple Inverse Kinematics controller for the manipulator has been developed. In addition, a simulation has been created showcasing the scenario and our implementation.

Index Terms—Automated Cable Crimping, UR5e Robotic Manipulator, Cable Detection and Handling

LIST OF ACRONYMS

HSV	Hue, Saturation, Value.
IK	Inverse Kinematics.
KD-tree	K-Dimensional tree.
MAE	Mean Absolute Error.
MSE	Mean Squared Error.
RF	Robot Frame coordinate system $[X, Y, Z]$ with origin in robot base.
RGB-D	Spectrum of red, green, blue and depth channels.
RMSE	Root Mean Squared Error.
TCP	Tool Center Point.
UR5e	Universal Robots robotic arm.
WF	World Frame coordinate system $[X, Y, Z]$.

I. DESCRIPTION OF THE INDUSTRIAL CHALLENGE

The industrial challenge we tried to solve was defined as "automating the installation of cables", which main focus was the detection and manipulation of flexible cable-like objects. We decided to approach this challenge from the angle of a crimping operation of an electrical cable, an operation that is still done by hand in many factories. The "automated" part of the challenge was covered by the use of a UR5e robotic manipulator, accompanied by a FESTO pneumatic gripper and a RealSense camera. More details about the specific hardware used will be covered in the Experimental Prototype section.

The most challenging part of the task is, besides the randomness of the position and configuration of the cable, the different possible colors, sizes, diameters and flexibilities, the different lighting conditions, and the different workspaces, to

correctly detect the cable as well as its end, and then pass the exact 3D position and orientation needed so that the robotic manipulator can grab it and transfer it to the crimping machine.

The crimping of cables is an important step in the construction process of a lot of different connectors, a repetitive task that uses a lot of man-hours for its completion, and sometimes also causes fatigue injuries. By introducing automation in the task, we eliminate these problems and free up resources that can be used in other areas of the factory. For this project, we contacted a major cable manufacturing company, confirming the importance and usefulness of automating this task.

II. STATE OF THE ART

[1] Early approaches to wire and cable detection, relied on traditional computer vision techniques. Kasturi et al. [2] introduced one of the first pipelines combining edge detection using the Steger algorithm [3] with a thresholded Hough transform [4]. This three stage structure edge extraction, line detection, and filtering became a common paradigm. Subsequent works refined these stages using background suppression, clustering, and morphological filtering [5].

More recent work has demonstrated that color-based methods remain effective in industrial robotic settings. Zhang et al. [6] proposed an HSV-based soft wire detection and 3D localization framework using contour filtering, geometric constraints, and depth reprojection for real-time obstacle avoidance in robot manipulators. Their results highlight the suitability of HSV representations for wire detection under controlled lighting and known object appearance.

State-of-the-art methods for deformable linear object perception, such as TrackDLO [7], model wires as continuous 3D entities and exploit temporal consistency for tracking. TrackDLO has shown strong robustness and repeatability across repeated trials and is therefore commonly used as a reference method in experimental evaluations.

In contrast, our HSV-based approach targets structured industrial environments with standardized setups and known wire appearance. While less general than learning-based methods, it offers low computational complexity, real-time performance, and straightforward integration with robotic manipulation pipelines.

III. APPROACH

We can partition the challenge into two parts, each with its own difficulties. We first had to detect a cable in an unknown

configuration and position inside our workspace, and then successfully grab it and move the end of the cable inside a crimping machine hole. We decided to use two different wire detection strategies, one that is considered the state of the art and one that is more simplistic and accessible.

To aid with the development of the challenge, a simulation of the task was firstly created, where we designed the workspace and the movement of the robot we will try to achieve. In the real setup, we decided to test at first the case where the cable is at a constant and elevated from the table configuration, in order to make its detection easier and safer for the manipulator in the case of collision with the table. Afterwards, we moved to the more useful case with the cable laying on top of the table. The model of a crimping hole to simulate the crimping machine and the elevated base for the cable has been created with the use of 3D printers.

A. Simulation

In the development cycle of the task at hand, a place to design and test some of the planned steps is crucial when the safety of equipment is important. Since we are dealing with equipment worth multiple thousands of euros, we created a simulation using the MuJoCo [8] advanced physics simulation to create the scenario we will try to reproduce in the real setup and ensure that the movement of the robotic components are safe.

We created a scene that mimics the real life testing conditions (Fig. 1), and is comprised of the same robotic manipulator, a realistic cable and a simulated crimping hole and elevated cable base. The camera component of this project is handled separately, and since it does not add any value to the simulation, it was not part of it. We mainly focused on the demonstration of the movement of the manipulator in the context of the cable and the rest of the scene. The controller used is a pure position inverse kinematics controller that runs a loop of iterative damped least squares and in each control step it computes the joint position targets and writes them directly to the virtual actuators (a kinematic position controller). The gripper control is also simple, performing a closing operation with a soft compliance to grab and hold the simulated cable during the entire movement. Since the workspace that we are working in is void of any singularities, and especially on a simple demonstration simulation, the position controller is enough to perform the task at hand, its limitations don't hinder the aim of this simulation, and we decided that a more sophisticated controller is not needed.

After we were satisfied with the behaviour of the simulated cable and settled on the movement of the manipulator and the complete scenario that we want to reproduce, we moved on to the real setup.

B. Robot control

In terms of the robot motion, the `ur_rtde` library [9] with its Python API has been used. The control system consist of two classes, `MotionPlanner` and `RobotControl`, handling motion planning and robot execution respectively. Both classes

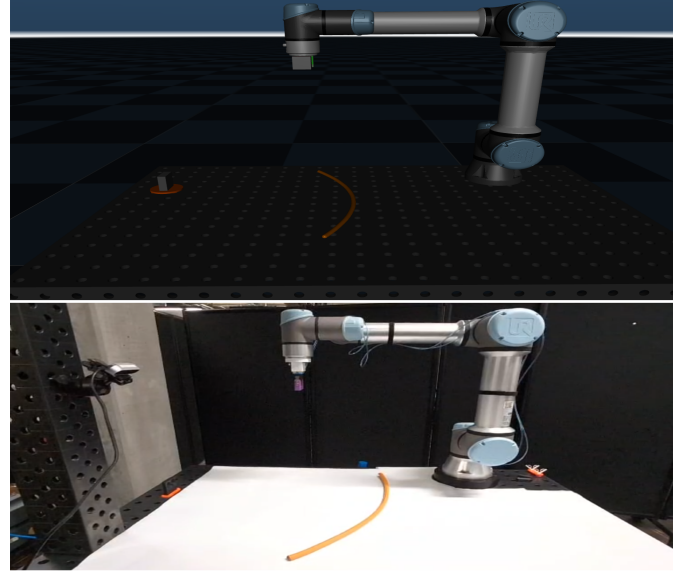


Fig. 1: Simulation to real system comparison.

maintain internal UR5e representation using `DHRobot` class from Robotics Toolbox [10] enabling visualization and inverse kinematics computations for the full kinematic chain.

`MotionPlanner` class converts given via points in the WF into joint space and plans trajectories according to specified timing or velocity, supporting profiles such as trapezoidal, quintic or linear with polynomial blend. Joint space trajectories support movement verification and allow partial execution through provided via point indexes. Visualization of static poses and complete movements is also implemented, helping to prevent potentially hazardous situations.

`RobotControl` class simplifies real robot operation by wrapping the `ur_rtde` library with convenient functions to execute planned joint space trajectories, primarily by using `servo_J` control. Movements can be interrupted at any time by pressing **Esc** key, adding additional safety in case of an unexpected behaviour. Additional functions for linear motion in WF were also implemented for tasks requiring slow, precise movements, such as inserting or grabbing cable at specified positions.

The main benefits of such implementation are that the TCP and robot base offsets are incorporated directly in planning, and base-mount twist is handled in control, enabling straight-forward usage and verification of the TCP position in WF.

Challenges in robot control included accounting for base mount, which introduces a twist of $-\frac{1}{8}\pi$ around the z -axis from WF to RF. Cable grabbing require a specific TCP orientation to avoid unnecessary cable twisting, but IK introduced undesired joint movements. This issue arises because the IK solver provides angles only within $\pm\pi$. For example a desired joint position of $\frac{10}{9} \cdot \pi$ while at $\frac{\pi}{2}$ returned $-\frac{8}{9} \cdot \pi$, even though it is within UR5e joint limits ($\pm\pi$). This problem was resolved by checking for discontinuities between via positions. The trajectory generation process is described in algorithm 1.

Algorithm 1 Trajectory Generation from Via Points

- 1: Obtain home configuration
 - 2: Receive sequence of via points in WF
 - 3: **for** each via point in sequence **do**
 - 4: Compute corresponding joint configuration using IK
 - 5: **end for**
 - 6: Optimize joint rotations between configurations
 - 7: Generate final trajectory
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C. Camera Calibration

To obtain the transformation between the Intel RealSense D415 camera and the UR5 robot base frame, we performed a standard camera-to-robot calibration using multiple ArUco [11] markers placed at known positions in the workspace. The 3D marker positions were measured in the robot base frame and detected in the camera frame using OpenCV [12]. A rigid transformation $T_{\text{cam} \rightarrow \text{base}}$ was then estimated from the resulting point correspondences using an SVD-based alignment method and validated by reprojection error analysis.

D. Wire Detection

1) *TrackDLO-Based Detection*: We initially adopted the TrackDLO framework [7] as a state-of-the-art solution for the perception of deformable linear objects, including wires, ropes, and cables. TrackDLO formulates wire perception as a spatio-temporal tracking problem and integrates a deep learning-based segmentation network with a graph-based centerline representation. The segmentation module produces dense pixel-level detections of the wire, which are subsequently organized into a structured graph capturing the topology and continuity of the object. By exploiting temporal consistency across consecutive frames, TrackDLO is able to generate a smooth and ordered centerline, estimate wire endpoints through graph traversal, and maintain stable tracking under moderate deformation and motion.

Using the official implementation, we extracted a dense set of 2D wire points, an ordered centerline representation, and wire endpoints, which were further reprojected into 3D using aligned RGB-D measurements. This pipeline enables accurate geometric reconstruction of the wire in the camera frame and supports downstream robotic tasks such as grasp planning and obstacle avoidance. In our experiments, TrackDLO demonstrated high robustness and repeatability across multiple trials, often producing consistent pick positions even under small variations in viewpoint and wire configuration.

Despite its strong performance, TrackDLO is computationally demanding due to its reliance on deep neural networks and graph-based optimization, making real-time execution challenging on limited hardware. In addition, we observed sensitivity to appearance variations. These limitations motivated the exploration of a complementary, lightweight alternative tailored to structured industrial environments. Using the official implementation, we extracted:

- a dense set of 2D wire points,

- a smooth ordered centerline,
- wire endpoints estimated using graph traversal,
- a 3D reconstruction based on aligned RGB-D data.

This method is accurate and robust but computationally heavy.

2) *Lightweight HSV-Based Detection*: To complement TrackDLO and provide a faster alternative, we designed a classical computer-vision pipeline implemented in Python and OpenCV. The method relies solely on color segmentation in the HSV space, followed by morphological filtering and skeletonization.

Given a color image I , we first convert it to the HSV color space and apply thresholding to obtain a binary mask:

$$M(u, v) = \begin{cases} 1, & \text{if } \text{HSV}(u, v) \in [\mathbf{l}_{\text{green}}, \mathbf{h}_{\text{green}}], \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

To remove noise and fill small gaps, a sequence of morphological operations is applied:

$$M' = \text{close}(\text{open}(M)). \quad (2)$$

A one-pixel-wide representation of the wire is then obtained using iterative skeletonization:

$$S = \text{skeleton}(M'). \quad (3)$$

Each skeleton pixel (u, v) is reprojected into 3D space using the depth value d , camera intrinsics (f_x, f_y, c_x, c_y) , and the camera-to-base transformation:

$$\mathbf{p}_{\text{cam}} = \begin{bmatrix} (u - c_x) \frac{d}{f_x} \\ (v - c_y) \frac{d}{f_y} \\ d \end{bmatrix}, \quad \mathbf{p}_{\text{base}} = T_{\text{cam} \rightarrow \text{base}} \begin{bmatrix} \mathbf{p}_{\text{cam}} \\ 1 \end{bmatrix}. \quad (4)$$

After collecting all valid 3D wire points $\{\mathbf{p}_i\}$, the wire endpoints are estimated using a farthest-point strategy implemented via a KD-tree search:

$$\mathbf{e}_1 = \arg \max_i \|\mathbf{p}_i - \mathbf{p}_0\|, \quad \mathbf{e}_2 = \arg \max_j \|\mathbf{p}_j - \mathbf{e}_1\|. \quad (5)$$

The wire direction vector is then computed as:

$$\mathbf{d} = \frac{\mathbf{e}_2 - \mathbf{e}_1}{\|\mathbf{e}_2 - \mathbf{e}_1\|}. \quad (6)$$

Pick positions are determined by identifying 3D points in the wire point cloud that lie at a fixed offset o from each endpoint along the wire direction. Specifically, points whose projection distance along $\mathbf{d}$ is within a small tolerance of the desired offset are selected and averaged:

$$\mathbf{p}_{\text{pick},1} \approx \mathbb{E}\{\mathbf{p}_i \mid |\langle \mathbf{p}_i - \mathbf{e}_1, \mathbf{d} \rangle - o| < \epsilon\}, \quad (7)$$

$$\mathbf{p}_{\text{pick},2} \approx \mathbb{E}\{\mathbf{p}_i \mid |\langle \mathbf{p}_i - \mathbf{e}_2, \mathbf{d} \rangle - (L - o)| < \epsilon\}, \quad (8)$$

where L is the estimated wire length and ϵ is a distance tolerance.

If insufficient point cloud samples are available at the desired offsets, the pick positions are computed analytically as:

$$\mathbf{p}_{\text{pick},1} = \mathbf{e}_1 + o \mathbf{d}, \quad \mathbf{p}_{\text{pick},2} = \mathbf{e}_2 - o \mathbf{d}. \quad (9)$$

E. Features of the HSV-Based Approach

The HSV-based wire detection approach is characterized by the following features:

- reliance on HSV color segmentation combined with morphological filtering and skeletonization,
- direct extraction of 3D wire points through RGB-D reprojection,
- straightforward integration with robotic manipulation pipelines.

The HSV-based method exhibited high sensitivity to lighting conditions and segmentation quality. However, due to its low computational complexity and minimal setup requirements, the HSV-based approach can be advantageous in structured environments with a standardized setup and known wire appearance, or in more flexible scenarios where lightweight and easily adaptable solutions are preferred.

IV. EXPERIMENTAL PROTOTYPE

The experimental prototype consists of a UR5e mounted on a welding table, an Intel® RealSense RGB-D camera and Festo® DHPS-16-A-NO pneumatic gripper with accessories.

To mount the gripper, a custom adapter was designed (Fig. 2b), along with claws for cable handling (Fig. 2a). The claws were designed to grasp cables from flat surfaces and provide high gripping force without flexing. Due to the limited 10 mm stroke, precise cable position is needed in order to successfully grab the cable. A scissor mechanism was considered but rejected, as plastic 3D printed version would likely introduce imprecise motion due to structural flexing during cable handling.

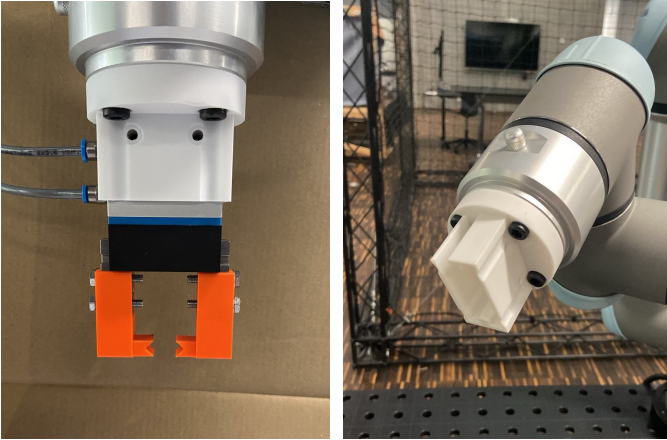


Fig. 2: End effector detail.

For testing purposes, the cable was initially fixed in a holder (Fig. 3b) to prevent collisions with the table. This constraint was removed once the system demonstrated reliable operation. In the prototype, the cable was assumed to lie flat on the table, with a fixed z-position used to avoid accidental table contact in the presence of sensor noise. In a real deployment, the table would be equipped with a deformable surface or

the gripper would incorporate compliance along the z-axis to prevent damage to the robot.

Insertion was performed into plastic mockup (Fig. 3a) of a manual crimping station inspired by CRIMPFIX ECO of Weidmüller (Fig. 4) with predefined target position, as the robot and station are considered a rigid, tightly coupled system with fixed relative positioning.



(a) Crimping station mockup. (b) Cable holder for testing.

Fig. 3: Testing equipment.



Fig. 4: Crimping Station of Inspiration

In order to test the limits of our implementation, two cables of different colors and diameters were tested: an orange and a green cable with a diameter of 13 mm and 8 mm respectively. As seen in Fig. 5 the output visual data of the specific camera used are noisy, preprocessed, and barely capable of recognizing the cable as it nearly blends into the background, reducing significantly the robustness of our implementation. The experimental results will be discussed in the next section of this report.

V. RESULTS

In this section we describe the performance of the two used methods as well as the procedure used to collect the test data.

A. Data collection

To compare the two employed methods, several tests were conducted. First, the wire detection algorithms were evaluated

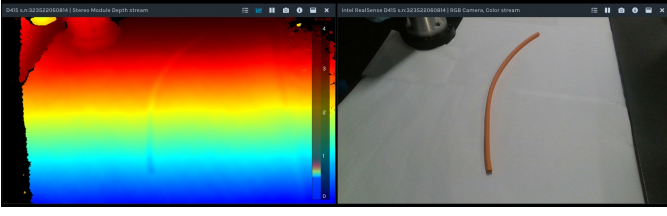


Fig. 5: Depth sensor output.

in isolation by performing 100 detection trials for each of 10 different wire positions. This procedure provided the mean and variance for each position, thereby quantifying the robustness and repeatability of the algorithms.

B. Statistical Analysis

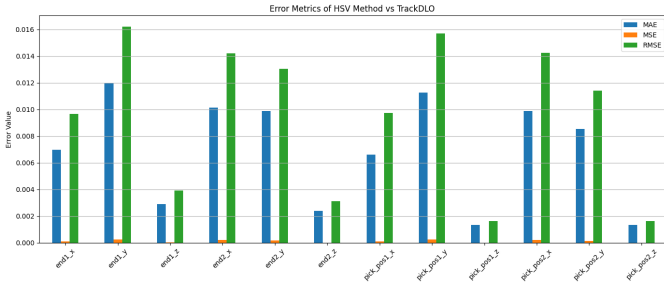


Fig. 6: Error metrics for the HSV-based method compared to TrackDLO.

1) *Quantitative Error Comparison:* The HSV-based wire detection method was compared to TrackDLO using MAE, MSE, and RMSE for wire endpoints and pick positions along the x , y , and z axes. As no absolute ground truth was available, TrackDLO was used as a reference method, and the reported metrics represent deviations relative to this reference. The HSV-based approach shows larger deviations, particularly in the x and y directions, while deviations along the z axis remain low. Overall, the results indicate that the HSV-based method provides acceptable consistency in structured environments, whereas TrackDLO offers more stable and repeatable detections across trials.

2) *Correlation between mean and variance:* For both methods, the dependence of detection precision on cable position was analyzed by computing the mean and variance for each position, including cable ends and grasping points.

The HSV-based method showed a weak negative correlation between mean position and variance (-0.18 in x , -0.11 in y). TrackDLO exhibited a similar correlation in y (-0.12) and almost no correlation in x (0.04). This suggests slightly lower variance for positions closer to the camera, except in the x -direction for TrackDLO, where variance remained nearly constant (Fig. 7).

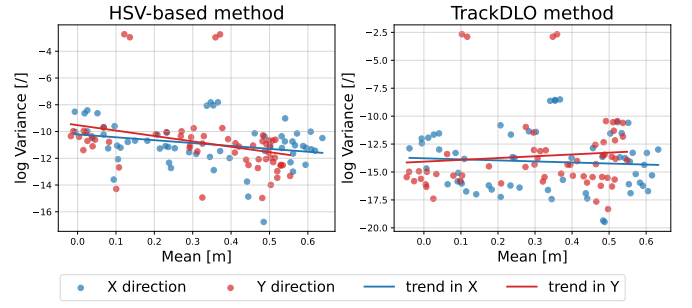


Fig. 7: Correlation trends between means and variances.

These results suggest that the HSV-based method exhibits higher variance in the x -direction, with no substantial difference in the y -direction. To verify this, two-sided F-tests were conducted. In the x -direction, the F-value was approximately 3.52 with $p = 2.2 \times 10^{-16}$, indicating that the HSV-based method has significantly higher variance. In the y -direction, $F \approx 0.94$ with $p \approx 0.01$, suggesting that TrackDLO has a slightly but statistically significant higher variance. But from variance per position in Fig. 8 can be seen that TrackDLO has consistently smaller variance (log ratio > 0) and for few exceptions (log ratio < 0).

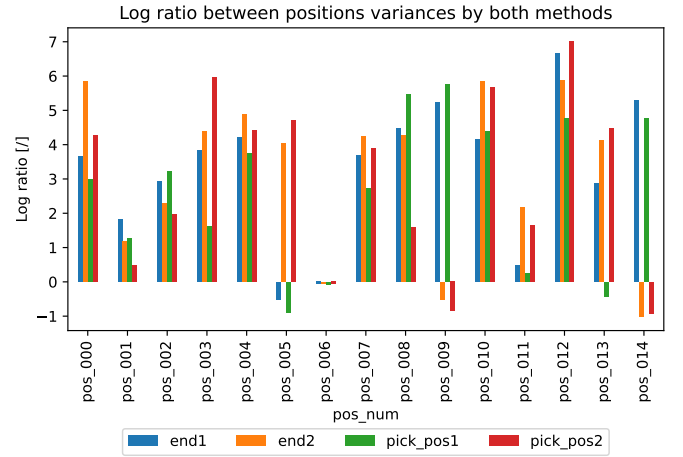


Fig. 8: Log ratio between HSV-based and TrackDLO.

3) *Data Distribution:* We performed a normality test for both the HSV-based method and the our TrackDLO implementation. We assessed whether each position's 2D point cloud followed a Gaussian normal distribution. For each of the 4 different positions (two ends of the cable and the two pick positions with 100 samples each), we checked the two coordinates separately and also tested the pair jointly, using standard normality checks (Shapiro–Wilk on the marginals) and a multivariate test (Henze–Zirkler), with the p-values adjusted for multiple comparisons. Using these criteria, only a small number of position sets were judged consistent with a bivariate normal model, while the majority showed statistically detectable differences from a normal distribution.

In some example diagnostics (Fig. 9) we can visually detect that the distributions are close to normal in the Q-Q plots, however formal statistical tests show that normality of the distribution cannot be assumed. Where non-normality was detected the probable causes were a few outlying points, heavier tails than a Gaussian, or slight skewness.

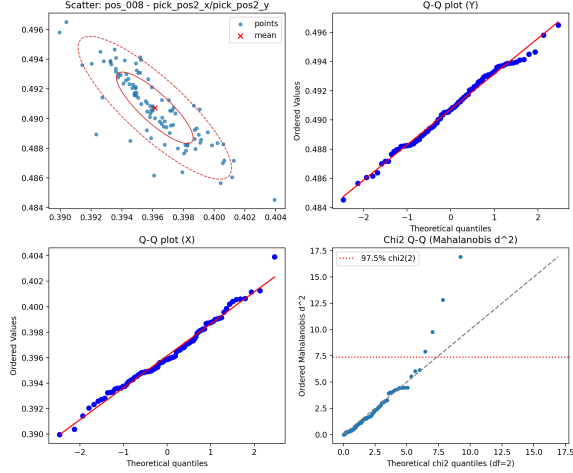


Fig. 9: Distribution analysis of HSV-based method of the pick position at one of the test positions of the orange cable.

C. Wire detection robustness

In practice, wire detection reliability was greatly affected by visual appearance and sensing conditions. TrackDLO was generally robust but showed instability in our setup and could not track the green cable consistently. It was mainly due to inconsistent feature extraction and sensitivity to color and texture changes. The HSV-based method showed steady performance under fixed lighting conditions, although it was very sensitive to changes in illumination. Variations in ambient light or time of day required us to manually adjust the HSV thresholds for reliable segmentation. Furthermore, the range of valid HSV parameters was narrow. Both methods were highly sensitive to small shifts in color representation that are probably exacerbated by noise and color inconsistencies in the camera output. These findings indicate that while HSV-based detection works fine in controlled settings, its reliability decreases under changing lighting conditions, unless we apply adaptive thresholding or color normalization, while the TrackDLO method is more unpredictable and in some cases fails completely.

D. Success rate

Talking about the success rate of our system, we defined as a successful attempt the case where we detected and grabbed the cable and successfully inserted it in the crimping hole. Additionally, in the testing process we only count as viable attempts, those where the cable is placed within the field of

view of the camera with both ends visible and on top of the table.

We performed 50 consecutive tests for each wire detection method for each cable color, with the exception of the TrackDLO method which failed to detect the green cable. The HSV thresholding method out of the 100 tests total, failed to produce a viable pick position 3 times, once with the orange cable and twice with the green. Incredibly, the TrackDLO method succeeded in all 50 of the tests with the orange cable.

TABLE I: Success rate

Cable Color	HSV Thresholding	TrackDLO
Orange	98%	100%
Green	96%	-

It is important to note that due to the nature of our approach, if the cable detection of the camera is correct and precise, then the manipulation part of the test and the insertion in the crimping hole are very unlikely to fail. Something that is also prevalent in the test results, where the only failures are due to wrong pick positions produced by the camera.

VI. CONCLUSION

To conclude this report, we successfully implemented an automated crimping procedure prototype, utilizing two cable detection methods. The performance of the system turned out to be quite high with a 100% success rate in one of the test cases. After investigating a more tolerable case with a wider orange cable, we stress-tested our approach using a smaller and harder to detect green cable where we found out the limits to our implementation of the TrackDLO method.

In the future, we would like to expand our setup with a real crimping machine in order to include the constraints a real machine brings to the problem. Additionally, including a more flexible cable will make the resulting system more close to some industrial use-cases and will force the approach to better implementations of TrackDLO or other state of the art methods, capable of handling detection of such sensitive cases.

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Individual contribution:

TABLE II: *Table of contributions*

Task description	Contributions
Simulation	Ioannis
Robot Control	Filip
Wire Detection	Vasileios
Data Gather	Vasileios, Filip, Ioannis
Data Analysis	Vasileios, Filip, Ioannis
Report Writing	Vasileios, Filip, Ioannis